IS6400 – Business Data Analytics (2026/27 Sem A)

*Time & classroom:	Offline Mode S01 & SA1: Mon. 15:00-17:50; LI-3508 S02 & SA2: Wed. 12:00-14:50; YEUNG-LT18 S61 & SP1: Wed. 19:00-21:50; CMC-M8051
Instructor:	**Junming Liu, Ph.D. AC3 6-277 (3442-9323)**
Email:	junmiliu@cityu.edu.hk
Office hour:	Junming: Mon. 12:00 – 14:00, by email appointment.
Co-Instructor (Grader):	SHUAI Yanqin, [contact removed] QIN Yaxue, [contact removed] Yu Shujing, [contact removed]

COURSE DESCRIPTION

The course aims to teach students the process, models, and tools for data analysis and analytics in business, such as in finance, marketing, etc. The course will teach students the practical skills to implement software packages (such as Python and GenAI) to tackle business data analysis problems for corporation manage, and decision making. On completion of the course students should be able to

1. understand the target and requirements of a selection of critical business data analysis problems.
2. manage the statistical techniques and machine learning models for data analytics.
3. Manage GenAI tools for BDA proposal and Python programming
4. implement the models into a software packages, such as Python, and adapt the models through programming capabilities.
5. analyze and interpret the outputs of models to support decision making in finance, marketing, accounting, etc.

READINGS:

Instructional Materials: Lecture slides, tutorial materials and other readings will be distributed throughout the semester.

Textbook (optional):

1. *Introduction to Data Mining*, Pearson, 2019. Pang-Ning Tan; Michael Steinbach; Anuj Karpatne; Vipin Kumar. ISBN-13: 978-0133128901 eTextbook Link
2. *Business Analytics: Data Analysis and Decision Making,* Albright, S.C. and Winston W.L., Cengage Learning, 2014.

ASSESSMENT:

Item#	Assessment	Weight
1	Class Participation	5%
2	Individual Assignments	25%
3	Group Project Briefing	5%
3	Group Project Presentation	15%
4	Group Project Report	10%
5	Exam	40%

COURSE SCHEDULE (*TENTATIVE*)

Week#	Lecture	Hands-on Tutorial	Tutor
1	Course Introduction	Software installation	Yaxue Qin
2	Predictive Analytics: Multivariate regression	Regression Analysis	Yaxue Qin
3	Data Exploration	Descriptive Analytics	Yaxue Qin
4	Feature Selection and Dimension Reduction	Principle Component Analysis	Yaxue Qin
5	Bisecting KMeans, DBSCAN	Clustering Methods	Yaxue Qin
6	Project Briefing		
7	Supervised Learning: Decision Trees, Ensemble Learning	Classification Methods	Yanqin Shuai
8	Time series analysis	Holt's Model, Arima	Yanqin Shuai
9	Deep Learning: ANN and RNN	Nonlinear Time Series	Yanqin Shuai
10	Deep Learning: Image Mining	Image Recognition	Yanqin Shuai
11	Model Assessment Competition: GenAI for BDA	Model Selection	Yanqin Shuai
12&13	Group Project Presentation		

Notes about grading:

1. Class Participation
In several classes that are chosen randomly, an in-class quiz (and signup sheet) will be conducted during the class. The portion of times that you conduct quiz correctly (and in class) will be used to assess your participation grade.

2. Individual Assignments
There are about eight assignments, which will be distributed later in the semester. Assignments are highly related to the lectures and the tutorials conducted in class. **Late submission will result in 20% deduction per day delay.**

3. Group Reports
The maximum number of students in a group is 5. The project presentations and report will be graded according to the quality of presentation and report. Detailed guidance about the group presentation and report will be discussed in the class. **Late submission for group report will result in 10% deduction per day delay.**

4. Final Exam
A final exam will be given at the end of the semester.